



Experiment 5

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Subject Name: Cloud Computing

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1. Aim: Configure Load Balancer with EC2 Instances.

2. Objective:

- To understand the concept and working of an AWS Load Balancer.
- To configure an Application Load Balancer with multiple EC2 instances.
- To distribute incoming traffic evenly across EC2 instances.
- To improve application availability and fault tolerance using load balancing.
- To verify load balancing by testing traffic distribution across instances.

3. Input / Apparatus Used:

- AWS account.
- Web browser.

4. Steps/Procedure:

1. Sign in to the AWS Management Console.
2. Search for **VPC** and open the VPC service.

Step 1: Launch Two EC2 Web Servers

3. Name: WebServer1
4. AMI: Amazon Linux 2
5. Instance type: t2.micro
6. Key pair: Create or select an existing key pair
7. Security group:
8. Allow SSH (22) from your IP
9. Allow HTTP (80) from 0.0.0.0/0
10. Launch the instance

Launch Second Instance

11. Name: WebServer2
12. Use the same configuration as WebServer1



13. Launch the instance
14. Wait until both instances are in Running state.
15. Note down the public IP addresses of both instances (for testing/verification).

Step 2: Create Application Load Balancer

16. In EC2 console, go to Load Balancers (left sidebar).
17. Click Create load balancer.
18. Select Application Load Balancer → Click Create.
19. Configure ALB Basics
20. Name: MyALB
21. Scheme: Internet-facing
22. IP address type: IPv4
23. Network Mapping
24. Select your VPC
25. Select at least two Availability Zones with public subnets

Step 3: Create Security Group for ALB

26. Create a new security group:
27. Name: alb-sg
28. Inbound rule:
29. Type: HTTP
30. Port: 80
31. Source: 0.0.0.0/0
32. Save the security group and attach it to the ALB.

Step 4: Create Target Group & Register Instances

33. Under Listeners and routing, click Create target group.
34. Target Group Configuration
35. Target type: Instances
36. Name: MyTargetGroup
37. Protocol: HTTP
38. Port: 80
39. VPC: Select your VPC
40. Health check path: / (default)
41. Click Next and register targets:
42. Select WebServer1
43. Select WebServer2
44. Click Include as pending below



45. Click Create target group

Step 5: Finish ALB Creation & Test

46. Return to the load balancer creation page.

47. In the target group dropdown, select MyTargetGroup.

48. Click Create load balancer.

49. Wait until the ALB state becomes Active.

Step 6: Verification

50. Copy the DNS name of the Application Load Balancer.

51. Paste it into a web browser.

52. Refresh the page multiple times.

53. Observe traffic being distributed between WebServer1 and WebServer2.

5. Result & Observations:

- Application Load Balancer configured successfully.
- EC2 instances registered in target group.
- Traffic distribution verified.

The screenshot displays the AWS Management Console interface for the 'Load balancers' section. On the left, a navigation pane shows categories like 'Elastic Block Store', 'Network & Security', 'Load Balancing', and 'Auto Scaling'. The 'Load Balancing' section is expanded, showing 'Load Balancers'. The main content area shows a table with one load balancer, 'myapplb', which is in an 'Active' state. Below the table, the 'Load balancer: myapplb' details are shown, including its type (Application), scheme (Internet-facing), VPC ID, and availability zones.

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
myapplb	Active	application	Internet-facing	IPv4	vpc-01a5923ab720bee79	2 Availability Zones

Load balancer: myapplb			
Details			
Load balancer type Application	Status Active	VPC vpc-01a5923ab720bee79	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z2STA26LKP4MIO	Availability Zones subnet-07ad5f1f70f5bab8d eu-north-1b (eu-n1-az2) subnet-0e37bc0e69c0f5586 eu-north-1a (eu-n1-az1)	Date created February 13, 2026, 14:46 (UTC+05:30)



6. Learning Outcome:

- Students will be able to configure a load balancer with multiple EC2 instances in AWS.
- Students will be able to register EC2 instances and configure health checks for traffic management.
- Students will be able to distribute and monitor incoming traffic across instances using a load balancer.
- Students will be able to implement high availability and fault tolerance for cloud applications.
- Students will be able to deploy scalable and reliable applications using AWS load balancing services.